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Pondering over the operation status and security issue of NB-IoT

Fan Zhou^{a,*}, Lei Li^b

^aUrban Vocational College of Sichuan ,

^bChina Mobile IOT Company Limited

Abstract

As a cellular communication technology, NB-IoT has become one of the development mainstreams of the Internet of Things due to its advantages, such as structure, operating costs, massive connections, extremely low power consumption, etc. NB-IoT was firstly put into the commercial use in the year of 2017. In this study, first, the features, development history and deployment status of NB-IoT are introduced. Then, the current security issues and challenges of NB-IoT are studied in the aspects of perception layer, network layer and application layer.

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1. Introduction

As the technologies of wireless connection, sensor design, chip manufacturing, storage, computing, networking and other technologies continually develop, the Internet of Things has grown from M2M communications to Internet of Things (IoT) which combines connections, big data, cloud computing, complex scenarios and intelligence. At present, in view of the intersection of the physical world and the virtual world, the intelligent connection of all things, such as people and people, people and things, things and things, will eventually become a reality. The era of the Internet of Things is approaching. In response, the cellular-based Narrow Band Internet of Things (NB-IoT) has emerged.

* Corresponding author. Tel.: +86-158-2863-6644

E-mail address: 563822964@qq.com

2. Why does the NB-IoT emerge?

The mobile communication networks are merely created for people's communication from 2G to 4G. However, with the advent of the Internet of Things era, it is necessary for mobile communication networks to proceed towards connecting "things". Based on this situation, 3GPP formulates the NB-IoT standard in Release 13 to meet the current requirements of the Internet of Things. Three kinds of cellular Internet of Things standards are defined: namely, EC-GSM, eMTC (LTE-M) and NB-IoT. GSM is the earliest wide-area M2M wireless connection technology, and C-GSM has enhanced its functions and competitiveness. Based on the evolution of LTE technology, LTE-M belongs to a subset of LTE. Despite of the close correlation between NB-IoT and LTE, it is an independent new air interface technology. The cellular Internet of Things NB-IoT achieves the aim of coverage enhancement, power consumption reduction, cost reduction and connections increase at the cost of speed and transmission delay. It is better to apply it in IoT devices that transmit small amounts of data, delay insensitively, and move at low speed, such as smart smoke detectors, shared bicycle, smart meter reading, smart manhole covers, which require the low amount of data that needs to be transmitted every day, allow a certain transmission delay and moves at low speed.

The cellular-based NB-IoT network consists of three deployment ways: in-band, guard band, and independent carrier. It can coexist with existing networks and be directly deployed on GSM, UMTS, or LTE networks, realizing smooth upgrades and reduction of deployment costs. Because of the low power consumption of NB-IoT, the standby time of NB-IoT terminal equipment is long, and the battery life can be as long as 10 years. In the meantime, due to the low receiving sensitivity of NB-IoT. The indoor cellular data connection coverage can be comprehensively acquired.

In addition, high-efficiency connection to terminal equipment can be achieved.

3. Development history and current status of NB-IoT

3.1. Development history of NB-IoT

Since NB-IoT is positioned as the prelude and foundation of 5G at the beginning of its design, its technical standards will evolve to the 5G era. Looking back at the evolution of NB-IoT technology, the 3GPP R13 version of NB-IoT is the basic one. R14 has increased the speed and enhanced multi-carrier functions while R15 has optimized the delay and reduced power consumption. Moreover, R16 has further improved performance, which can be accessed to the 5G core network. Additionally, the R17 version of the enhanced NB-IoT has been officially approved. NB-IoT has advanced from the R13 and R14 versions in the 4G era to R15, R16, and R17 in the 5G era and is an important component of 5G.

In comparison with the traditional Internet of Things technology, NB-IOT is characterized by wide coverage, massive connections, low speed, low cost, low power consumption, and excellent architecture. Therefore, it comprehensively promotes the coverage and stability of the Internet of Things, and greatly meets the diversified needs and the development needs of ultra-long-distance communication technology. It is expected that Internet of Things applications such as remote meter reading, asset tracking, smart transportation, smart buildings, smart cities, industrial Internet of Things, wearable devices, etc. are booming.

3.2. Current Status of NB-IoT Deployment

In November 2019, the number of global NB-IoT connections reached around 70 million. In January 2020, the number of global connections topped 100 million. In February 2020, the number of connections in China alone exceeded 100 million. But nowadays, the number of global connections has increased to 140 million! The number of NB-IoT connections has doubled within merely one year, and the industry has shown explosive growth. According to GSMA statistics, as of September 2020, the number of overseas cellular LPWA connections surpassed 40 million, an increase of 148% over the same period last year. More than 20 large-particle NB-IoT projects in over 10 overseas countries are being deployed, invited or prepared. In terms of networks, 101 NB-IoT networks have been deployed globally. On top of that, NB-IoT has been activated by mainstream global operators. As of the end of August 2020, more than 700,000 NB-IoT base stations have been established, which have promoted the continuous coverage in

areas beyond the township across the nation. It will lay a good foundation for the development of various applications.

4. NB-IoT Security

4.1. Features of NB-IoT Security

Compared with the security issues of traditional Internet of Things, there are many similarities and some differences related to those of NB-IOT. The NB-IoT's security issues are mainly involved in the diversity of perception information, the heterogeneity of the network environment and the complexity of application requirements. Therefore, it presents the characteristics of the large-scale of the network, the large amount of data processing, and the complex decision-making control. The perception layer, network layer, and application layer are analyzed as below.

4.2. Security Issue and Thinking of Perception Layer

The perception layer, which is located at the bottom of the narrowband Internet of Things, is the source of its original data. The quality of the data gathered at the perception layer has a direct effect on the accuracy of the back-end data processing and correct implementation of the relevant control. When the narrowband Internet of Things collects data at the perception layer, its information transmission method mainly adopts wireless network transmission. Because of the exposed feature, wireless signals are easy to be illegally monitored, stolen and maliciously interfered. In addition, considering that some devices and nodes of the NB-IoT must be deployed in harsh, unattended environments, complicated and dangerous operations must be completed through remote control, which creates opportunities for attackers. Attackers can directly contact these devices to do the damage on the devices or the sensors they carry, perform node capture attacks and information tampering. Worse still, they forge or illegally manipulate the sensor communication protocols by deciphering them. Automatic identification technologies such as radio frequency identification RFID are the most commonly-used data collection methods for the perception layer. Following this, the paper will illustrate possible security risks by citing an RFID system as an instance. The RFID system consists of a transponder, reader, antenna, and application software system.

For example, in the RFID system, the data which is being transmitted or stored in the tag, reader or back-end system is subject to be assaulted at any time. At the same time, an attacker can read the data of tag through the reader or other means. RFID tags may be illegally utilized to track and violate others' location privacy. It may even be used by commercial espionage to modify or rewrite item information. When the reader and tags communicate via radio waves, an attacker may intercept data, block, deceive data communication, or even send data using illegal tags.

In terms of the attack methods mentioned above, cryptographic algorithms such as data encryption, identity authentication, and integrity verification can be adopted to prevent. The key management mechanism can be strengthened according to actual needs within the sensor, and temporary session keys can be established during communication. As for some key nodes or nodes that are easily attacked, intrusion detection, alarm, blockage and self-destruct procedures can be specially strengthened. If the node is found to leave the original application, initiates the alarm, blockade or self-destruct procedures should be started.

4.3. Security Issue and Thoughts on the Network Layer

The network layer has a powerful link function, which connects the perception layer and the application layer. It is mainly used to transmit the information collected by the perception layer to the information processing layer in safe and reliable manners. Then, it will perform information processing according to different application requirements.

The wireless communication is used by the perception layer and the network layer of NB-IoT. However, the inherent fragility of the wireless network will bring potential risks to the system. Attackers can carry out malicious interference or even interrupt communication by transmitting radio signals. In addition, one sector of NB-IoT can support nearly 100,000 terminal connections. If these massive, real-time, large-capacity connections cannot be

efficiently authenticated and controlled, criminals can inject false information by taking advantage of malicious nodes, or even launch Dos and DDos attacks by using the controlled node, further affecting the performance of the system.

In order to cope with the aforesaid network attacks, end-to-end encryption and node-to-node encryption can be strengthened. Furthermore, asymmetric cryptographic algorithms can be adopted to improve key management schemes and protocols and provide integrity and confidentiality protection for NB-IoT data transmission. In the case of Dos and DDos attacks, it is necessary to strengthen the detection and prevention of Dos and DDos. Illegal information injected by malicious nodes can be detected by setting up contour thresholds for certain NB-IoT nodes for the purpose of establishing a complete intrusion detection and protection mechanism.

4.4. Security Issues and Thinking on Application Layer

The application layer is a comprehensive or specific application program with individual characteristics. Typical security issues include privacy protection and intellectual property protection and other aspects. For instance, mobile users need to be legally aware of their location information, but they are worried that their information will be plundered by illegal users. Users need to use a certain business legally, but they do not want others to know that they are using a certain business. Many businesses require anonymity, such as online voting.

There is an urgent need for information leakage, intellectual property rights, and privacy issues to apply more advanced cryptographic techniques, such as access control, anonymous signature, anonymous authentication, cipher text verification, treason tracking, digital watermarking and fingerprint technology, etc.

With regard to the vulnerabilities in lease registration management, leased users shall be controlled strictly by following regulations. In addition to valid credit cards, identity and fingerprint entry can be added. However, as far as enterprise user information security management is concerned, it is better to be managed by a unified user information security platform established by the enterprise.

5. Conclusion

With the arrival of the NB-IoT era, the security issue are not only related to the development of NB-IoT technology, but also related to the privacy and interests of network operators and each user. This paper presents an analysis of the current security and challenges confronted by the NB-IoT in the aspects of the perception layer, network layer and application layer. Security is the top priority of NB-IoT. There are still many problems to be solved for NB-IoT in the future:

(1) Strengthen the reliability of the NB-IoT system further, including the network infrastructure of the operators and its own anti-attack capabilities.

(2) NB-IoT applications depend on the operator's network, so operators shall work out a series of secure access standards in order to maintain the NB-IoT security system, such as access permission, real-name registration, NB-IoT terminal authentication and accountability mechanism, and equipment positioning, etc.

(3) Cloud computing is one of the most critical technologies related to the Internet of Things. Although the "cloud" own tremendous resources, its tenancy registration management mechanism has certain security flaws, which means that it cannot distinguish the specific identities of users. Thus, there is a risk of resource abuse. Meanwhile, the cyber crime will be prevailing by taking advantage of this convenient channel. In addition, there are risks in the security management of enterprise users. Currently, since as the tenants, enterprises are in charge of related matters concerning network information security, some issues such as security auditing, security assessment and certification will emerge. Therefore, a unified platform to manage enterprise user information security are urgently needed.

References

1. China Industry Research Network. Research on the Development Trend of 2016 Internet of Things [EB/OL]. (2016-12-15) [2017-04-25]. <http://bg.qianzhan.com/report/detail/678e0de2511a4c18.html?bdqztd>.
2. D.L.You. Development Trend of the Internet of Things in China[J].Wireless Internet Technology,2017,(8): 95-96.

3. Research Institute of Telecommunications Under Ministry of Industry and Information Technology. In 2015, the scale of China's Internet of Things industry will exceed 700 billion yuan [EB/OL]. (2014-06-20) [2017-04-25].<http://news.jc001.cn/14/0620/808131.html>.
4. Q.Y.Yu . An Overview of the Concept and Technology Application of the Internet of Things[J]. Digital World, 2017, 276
5. Q.Gan/Text. Launch of Made in China "Internet of Things"[J]. China Automatic Identification Technology, 2017, (4):42-45
6. F. Wei: Thoughts on the Development and Construction of the Internet of Things in China[J]. China Science and Technology Investment, 2010